

## Activity Diagrams: - A Quick Overview

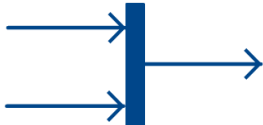
What is an Activity diagram?

A UML activity diagram helps to visualize a certain **use case at a more detailed level**. It is a behavioural diagram that **illustrates the flow of activities through a system**. Activity diagrams can also be used to depict a flow of **events in a business process**.

### Activity Diagram Symbols

UML has specified a set of symbols and rules for drawing activity diagrams. Following are the commonly used activity diagram symbols with explanations.

| Symbol                                                                              | Name                       | Use                                                                      |
|-------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------|
|    | Start/ Initial Node        | Used to represent the starting point or the initial state of an activity |
|  | Activity / Action State    | Used to represent the activities of the process                          |
|  | Action                     | Used to represent the executable sub-areas of an activity                |
|  | Control Flow / Edge        | Used to represent the flow of control from one action to the other       |
|  | Object Flow / Control Edge | Used to represent the path of objects moving through the activity        |
|  | Activity Final Node        | Used to mark the end of all control flows within the activity            |

|                                                                                     |                 |                                                                                  |
|-------------------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------|
|    | Flow Final Node | Used to mark the end of a single control flow                                    |
|    | Decision Node   | Used to represent a conditional branch point with one input and multiple outputs |
|    | Merge Node      | Used to represent the merging of flows. It has several inputs, but one output.   |
|    | Fork            | Used to represent a flow that may branch into two or more parallel flows         |
|  | Merge           | Used to represent two inputs that merge into one output                          |
|  | Signal Sending  | Used to represent the action of sending a signal to an accepting activity        |
|  | Signal Receipt  | Used to represent that the signal is received                                    |
|  | Note/ Comment   | Used to add relevant comments to elements                                        |

## Example activity diagrams

**The scenario:** - A customer needs a personal loan and approaches Peninsula Bank to apply for it. An application form is completed and handed in to a consultant. The consultant processes the application and sends it to the Credit Department for assessment. On its return the consultant lets the customer know the result of the application. Draw an activity diagram for a loan application

(10)

### Sample Answer with and without Swimlanes

